4.E The Brown Representability Theorem

In Theorem 4.58 in §4.3 we showed that Ω-spectra define cohomology theories, and now we will prove the converse statement that all cohomology theories on the CW category arise in this way from Ω-spectra.

Theorem 4E.1. *Every reduced cohomology theory on the category of basepointed CW complexes and basepoint-preserving maps has the form $h^n(X) = \langle X, K_n \rangle$ for some Ω-spectrum $\{K_n\}$.*

We will also see that the spaces K_n are unique up to homotopy equivalence.

This theorem gives another proof that ordinary cohomology is representable as maps into Eilenberg–MacLane spaces, since for the spaces K_n in an Ω-spectrum representing $\widetilde{H}^*(-;G)$ we have $\pi_i(K_n) = \langle S^i, K_n \rangle = \widetilde{H}^n(S^i;R)$, so K_n is a $K(G,n)$.

Before getting into the proof of the theorem let us observe that cofibration sequences, as constructed in §4.3, allow us to recast the definition of a reduced cohomology theory in a slightly more concise form: A reduced cohomology theory on the category $\mathcal{C}$ whose objects are CW complexes with a chosen basepoint 0-cell and whose morphisms are basepoint-preserving maps is a sequence of functors h^n, $n \in \mathbb{Z}$, from $\mathcal{C}$ to abelian groups, together with natural isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$ for all X in $\mathcal{C}$, such that the following axioms hold for each h^n:

(i) If $f \simeq g : X \to Y$ in the basepointed sense, then $f^* = g^* : h^n(Y) \to h^n(X)$.

(ii) For each inclusion $A \hookrightarrow X$ in $\mathcal{C}$ the sequence $h^n(X/A) \to h^n(X) \to h^n(A)$ is exact.

(iii) For a wedge sum $X = \bigvee_\alpha X_\alpha$ with inclusions $i_\alpha : X_\alpha \hookrightarrow X$, the product map $\prod_\alpha i_\alpha^* : h^n(X) \to \prod_\alpha h^n(X_\alpha)$ is an isomorphism.

To see that these axioms suffice to define a cohomology theory, the main thing to note is that the cofibration sequence $A \to X \to X/A \to \Sigma A \to \cdots$ allows us to construct the long exact sequence of a pair, just as we did in the case of the functors $h^n(X) = \langle X, K_n \rangle$. In the converse direction, if we have natural long exact sequences of pairs, then by applying these to pairs of the form (CX, X) we get natural isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$. Note that these natural isomorphisms coming from coboundary maps of pairs (CX, X) uniquely determine the coboundary maps for all pairs (X, A) via the diagram at the right, where the maps from $h^n(A)$ are coboundary maps of pairs and the diagram commutes by naturality of these coboundary maps. The isomorphism comes from a deformation retraction of CX onto CA. It is easy to check that these processes for converting one definition of a cohomology theory into the other are inverses of each other.

$$\begin{tikzcd}
h^n(A) \arrow[r] \arrow[d] \arrow[dr] & h^{n+1}(X/A) \\
h^{n+1}(CA/A) & h^{n+1}(CX/A) \arrow[l, "\approx"'] \arrow[u]
\end{tikzcd}$$

Most of the work in representing cohomology theories by Ω-spectra will be in realizing a single functor h^n of a cohomology theory as $\langle -, K_n \rangle$ for some space K_n.

So let us consider what properties the functor $h(X) = \langle X, K\rangle$ has, where K is a fixed space with basepoint. First of all, it is a contravariant functor from the category of basepointed CW complexes to the category of pointed sets, that is, sets with a distinguished element, the homotopy class of the constant map in the present case. Morphisms in the category of pointed sets are maps preserving the distinguished element. We have already seen in §4.3 that $h(X)$ satisfies the three axioms (i)–(iii). A further property is the following **Mayer–Vietoris axiom**:

- Suppose the CW complex X is the union of subcomplexes A and B containing the basepoint. Then if $a \in h(A)$ and $b \in h(B)$ restrict to the same element of $h(A \cap B)$, there exists an element $x \in h(X)$ whose restrictions to A and B are the given elements a and b.

Here and in what follows we use the term 'restriction' to mean the map induced by inclusion. In the case that $h(X) = \langle X, K\rangle$, this axiom is an immediate consequence of the homotopy extension property. The functors h^n in any cohomology theory also satisfy this axiom since there are Mayer–Vietoris exact sequences in any cohomology theory, as we observed in §2.3 in the analogous setting of homology theories.

Theorem 4E.2. *If h is a contravariant functor from the category of connected basepointed CW complexes to the category of pointed sets, satisfying the homotopy axiom (i), the Mayer–Vietoris axiom, and the wedge axiom (iii), then there exists a connected CW complex K and an element $u \in h(K)$ such that the transformation $T_u : \langle X, K\rangle \to h(X)$, $T_u(f) = f^*(u)$, is a bijection for all X.*

Such a pair (K, u) is called **universal** for the functor h. It is automatic from the definition that the space K in a universal pair (K, u) is unique up to homotopy equivalence. For if (K', u') is also universal for h, then, using the notation $f : (K, u) \to (K', u')$ to mean $f : K \to K'$ with $f^*(u') = u$, universality implies that there are maps $f : (K, u) \to (K', u')$ and $g : (K', u') \to (K, u)$ that are unique up to homotopy. Likewise the compositions $gf : (K, u) \to (K, u)$ and $fg : (K', u') \to (K', u')$ are unique up to homotopy, hence are homotopic to the identity maps.

Before starting the proof of this theorem we make two preliminary comments on the axioms.

(1) The wedge axiom implies that $h(\mathit{point})$ is trivial. To see this, just use the fact that for any X we have $X \vee \mathit{point} = X$, so the map $h(X) \times h(\mathit{point}) \to h(X)$ induced by inclusion of the first summand is a bijection, but this map is the projection $(a, b) \mapsto a$, hence $h(\mathit{point})$ must have only one element.

(2) Axioms (i), (iii), and the Mayer–Vietoris axiom imply axiom (ii). Namely, (ii) is equivalent to exactness of $h(A) \leftarrow h(X) \leftarrow h(X \cup CA)$, where CA is the reduced cone since we are in the basepointed category. The inclusion Im $\subset$ Ker holds since the composition $A \to X \cup CA$ is nullhomotopic, so the induced map factors through $h(\mathit{point}) = 0$.

To obtain the opposite inclusion Ker $\subset$ Im, decompose $X \cup CA$ into two subspaces Y and Z by cutting along a copy of A halfway up the cone CA, so Y is a smaller copy of CA and Z is the reduced mapping cylinder of the inclusion $A \hookrightarrow X$. Given an element $x \in h(X)$, this extends to an element $z \in h(Z)$ since Z deformation retracts to X. If x restricts to the trivial element of $h(A)$, then z restricts to the trivial element of $h(Y \cap Z)$. The latter element extends to the trivial element of $h(Y)$, so the Mayer–Vietoris axiom implies there is an element of $h(X \cup CA)$ restricting to z in $h(Z)$ and hence to x in $h(X)$.

The bulk of the proof of the theorem will be contained in two lemmas. To state the first, consider pairs (K,u) with K a basepointed connected CW complex and $u \in h(K)$, where h satisfies the hypotheses of the theorem. Call such a pair (K,u) ***n*-universal** if the map $T_u : \pi_i(K) \to h(S^i)$, $T_u(f) = f^*(u)$, is surjective for $i \le n$ and has trivial kernel for $i < n$. Note that having a trivial kernel may not be the same as being injective since we are not dealing with homomorphisms of groups here. Call (K,u) π_***-universal** if it is n-universal for all n.

Lemma 4E.3. *Given any pair (Z,z) with Z a connected CW complex and $z \in h(Z)$, there exists a π_*-universal pair (K,u) with Z a subcomplex of K and $u \,|\, Z = z$.*

Proof: We construct K from Z by an inductive process of attaching cells. To begin, let $K_1 = Z \bigvee_\alpha S^1_\alpha$ where α ranges over the elements of $h(S^1)$. By the wedge axiom there exists $u_1 \in h(K_1)$ with $u_1 \,|\, Z = z$ and $u_1 \,|\, S^1_\alpha = \alpha$, so (K_1, u_1) is 1-universal.

For the inductive step, suppose we have already constructed (K_n, u_n) with u_n n-universal, $Z \subset K_n$, and $u_n \,|\, Z = z$. Represent each element α in the kernel of $T_{u_n} : \pi_n(K_n) \to h(S^n)$ by a map $f_\alpha : S^n \to K_n$. Let $f = \bigvee_\alpha f_\alpha : \bigvee_\alpha S^n_\alpha \to K_n$. The reduced mapping cylinder M_f deformation retracts to K_n, so we can regard u_n as an element of $h(M_f)$, and this element restricts to the trivial element of $h(\bigvee_\alpha S^n_\alpha)$ by the definition of f. The exactness property of h then implies that for the reduced mapping cone $C_f = M_f / \bigvee_\alpha S^n_\alpha$ there is an element $w \in h(C_f)$ restricting to u_n on K_n. Note that C_f is obtained from K_n by attaching cells e^{n+1}_α by the maps f_α. To finish the construction of K_{n+1}, set $K_{n+1} = C_f \bigvee_\beta S^{n+1}_\beta$ where β ranges over $h(S^{n+1})$. By the wedge axiom again, there exists $u_{n+1} \in h(K_{n+1})$ restricting to w on C_f and β on S^{n+1}_β.

To see that (K_{n+1}, u_{n+1}) is $(n+1)$-universal, consider the commutative diagram at the right. Since K_{n+1} is obtained from K_n by attaching $(n+1)$-cells, the upper map is an isomorphism for $i < n$ and a surjection for $i = n$. By induction the map T_{u_n} has trivial kernel for $i < n$ and is surjective for $i \le n$, so the same is true for $T_{u_{n+1}}$. The kernel of $T_{u_{n+1}}$ is trivial for $i = n$ since an element of this kernel pulls back to Ker $T_n \subset \pi_n(K_n)$, by surjectivity of the upper map when $i = n$, and we attached cells to K_n by maps representing all elements of Ker T_n. And lastly, $T_{u_{n+1}}$ is surjective for $i = n+1$ by construction.

$$\begin{tikzcd} \pi_i(K_n) \arrow[rr] \arrow[dr, "T_{u_n}"'] & & \pi_i(K_{n+1}) \arrow[dl, "T_{u_{n+1}}"] \\ & h(S^i) & \end{tikzcd}$$

Now let $K = \bigcup_n K_n$. We apply a mapping telescope argument as in the proofs of Lemma 2.34 and Theorem 3F.8 to show there is an element $u \in h(K)$ restricting to u_n on K_n, for all n. The mapping telescope of the inclusions $K_1 \hookrightarrow K_2 \hookrightarrow \cdots$ is the subcomplex $T = \bigcup_i K_i \times [i, i+1]$ of $K \times [1, \infty)$. We take '$\times$' to be the reduced product here, with *basepoint* $\times$ *interval* collapsed to a point. The natural projection $T \to K$ is a homotopy equivalence since $K \times [1, \infty)$ deformation retracts onto T, as we showed in the proof of Lemma 2.34. Let $A \subset T$ be the union of the subcomplexes $K_i \times [i, i+1]$ for i odd and let B be the corresponding union for i even. Thus $A \cup B = T$, $A \cap B = \bigvee_i K_i$, $A \simeq \bigvee_i K_{2i-1}$, and $B \simeq \bigvee_i K_{2i}$. By the wedge axiom there exist $a \in h(A)$ and $b \in h(B)$ restricting to u_i on each K_i. Then using the fact that $u_{i+1} \,|\, K_i = u_i$, the Mayer–Vietoris axiom implies that a and b are the restrictions of an element $t \in h(T)$. Under the isomorphism $h(T) \approx h(K)$, t corresponds to an element $u \in h(K)$ restricting to u_n on K_n for all n.

To verify that (K, u) is π_*-universal we use the commutative diagram at the right. For $n > i + 1$ the upper map is an isomorphism and T_{u_n} is surjective with trivial kernel, so the same is true of T_u. □

$$\begin{tikzcd} \pi_i(K_n) \arrow[rr] \arrow[dr, "T_{u_n}"'] & & \pi_i(K) \arrow[dl, "T_u"] \\ & h(S^i) & \end{tikzcd}$$

Lemma 4E.4. *Let (K, u) be a π_*-universal pair and let (X, A) be a basepointed CW pair. Then for each $x \in h(X)$ and each map $f : A \to K$ with $f^*(u) = x \,|\, A$ there exists a map $g : X \to K$ extending f with $g^*(u) = x$.*

Schematically, this is saying that the diagonal arrow in the diagram at the right always exists, where the map i is inclusion.

$$\begin{tikzcd} (A, a) \arrow[r, "f"] \arrow[d, "i"'] & (K, u) \\ (X, x) \arrow[ur, "g"'] & \end{tikzcd}$$

Proof: Replacing K by the reduced mapping cylinder of f reduces us to the case that f is the inclusion of a subcomplex. Let Z be the union of X and K with the two copies of A identified. By the Mayer–Vietoris axiom, there exists $z \in h(Z)$ with $z \,|\, X = x$ and $z \,|\, K = u$. By the previous lemma, we can embed (Z, z) in a π_*-universal pair (K', u'). The inclusion $(K, u) \hookrightarrow (K', u')$ induces an isomorphism on homotopy groups since both u and u' are π_*-universal, so K' deformation retracts onto K. This deformation retraction induces a homotopy rel A of the inclusion $X \hookrightarrow K'$ to a map $g : X \to K$. The relation $g^*(u) = x$ holds since $u' \,|\, K = u$ and $u' \,|\, X = x$. □

Proof of Theorem 4E.2: It suffices to show that a π_*-universal pair (K, u) is universal. Applying the preceding lemma with A a point shows that $T_u : \langle X, K \rangle \to h(X)$ is surjective. To show injectivity, suppose $T_u(f_0) = T_u(f_1)$, that is, $f_0^*(u) = f_1^*(u)$. We apply the preceding lemma with $(X \times I, X \times \partial I)$ playing the role of (X, A), using the maps f_0 and f_1 on $X \times \partial I$ and taking x to be $p^* f_0^*(u) = p^* f_1^*(u)$ where p is the projection $X \times I \to X$. Here $X \times I$ should be the reduced product, with *basepoint* $\times I$ collapsed to a point. The lemma then gives a homotopy from f_0 to f_1. □

Proof of Theorem 4E.1: Since suspension is an isomorphism in any reduced cohomology theory, and the suspension of any CW complex is connected, it suffices to restrict attention to connected CW complexes. Each functor h^n satisfies the homotopy, wedge, and Mayer–Vietoris axioms, as we noted earlier, so the preceding theorem gives CW complexes K_n with $h^n(X) = \langle X, K_n \rangle$. It remains to show that the natural isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$ correspond to weak homotopy equivalences $K_n \to \Omega K_{n+1}$. The natural isomorphism $h^n(X) \approx h^{n+1}(\Sigma X)$ corresponds to a natural bijection $\langle X, K_n \rangle \approx \langle \Sigma X, K_{n+1} \rangle = \langle X, \Omega K_{n+1} \rangle$ which we call Φ. The naturality of this bijection gives, for any map $f : X \to K_n$, a commutative diagram as at the right. Let $\varepsilon_n = \Phi(\mathbb{1}) : K_n \to \Omega K_{n+1}$. Then using commutativity we have $\Phi(f) = \Phi f^*(\mathbb{1}) = f^* \Phi(\mathbb{1}) = f^*(\varepsilon_n) = \varepsilon_n f$, which says that the map $\Phi : \langle X, K_n \rangle \to \langle X, \Omega K_{n+1} \rangle$ is composition with ε_n. Since Φ is a bijection, if we take X to be S^i, we see that ε_n induces an isomorphism on π_i for all i, so ε_n is a weak homotopy equivalence and we have an Ω-spectrum.

$$\begin{array}{ccc} \langle K_n, K_n \rangle & \xrightarrow{f^*} & \langle X, K_n \rangle \\ \downarrow{\scriptstyle \Phi} & & \downarrow{\scriptstyle \Phi} \\ \langle K_n, \Omega K_{n+1} \rangle & \xrightarrow{f^*} & \langle X, \Omega K_{n+1} \rangle \end{array}$$

There is one final thing to verify, that the bijection $h^n(X) = \langle X, K_n \rangle$ is a group isomorphism, where $\langle X, K_n \rangle$ has the group structure that comes from identifying it with $\langle X, \Omega K_{n+1} \rangle = \langle \Sigma X, K_{n+1} \rangle$. Via the natural isomorphism $h^n(X) \approx h^{n+1}(\Sigma X)$ this is equivalent to showing the bijection $h^{n+1}(\Sigma X) = \langle \Sigma X, K_{n+1} \rangle$ preserves group structure. For maps $f, g : \Sigma X \to K$, the relation $T_u(f+g) = T_u(f) + T_u(g)$ means $(f+g)^*(u) = f^*(u) + g^*(u)$, and this holds since $(f+g)^* = f^* + g^* : h(K) \to h(\Sigma X)$ by Lemma 4.60. ◻

4.F Spectra and Homology Theories

We have seen in §4.3 and the preceding section how cohomology theories have a homotopy-theoretic interpretation in terms of Ω-spectra, and it is natural to look for a corresponding description of homology theories. In this case we do not already have a homotopy-theoretic description of ordinary homology to serve as a starting point. But there is another homology theory we have encountered which does have a very homotopy-theoretic flavor:

Proposition 4F.1. *Stable homotopy groups $\pi_n^s(X)$ define a reduced homology theory on the category of basepointed CW complexes and basepoint-preserving maps.*

Proof: In the preceding section we reformulated the axioms for a cohomology theory so that the exactness axiom asserts just the exactness of $h^n(X/A) \to h^n(X) \to h^n(A)$ for CW pairs (X, A). In order to derive long exact sequences, the reformulated axioms